

CROIP - Approach 1: Federated Model

Central Revenue Oversight and Integration Platform. Ratepayers deal with concessionaires; CROIP provides oversight and integration.

1. Overview

CROIP sits above independent concessionaire platforms as an oversight and integration hub. Each concessionaire keeps its own full stack and operational database. CROIP connects to all of them through a National Integration Layer (standard APIs, data standards, event streaming, VPN/TLS) and gives government a national view of revenue.

2. Who does what

Party	Role in this model
Ratepayer	Deals directly with their concessionaire for registration, billing and payment. Does not interact with CROIP.
Concessionaire	Runs the full stack: taxpayer registration, assessment and billing, collections (web, mobile, POS), field agents, customer service, local reporting.
CROIP	Oversight and integration only: transaction visibility, reconciliation, national reporting, fraud monitoring, audit, identity and reference data.
Payment channels	Banks, mobile money, cards, POS, web, USSD and agents connect to each concessionaire separately. Each concessionaire owns its own collections.

3. CROIP core components

- **Transaction Hub:** collects all concessionaire transactions in real or near real time.
- **Reconciliation Engine:** matches assessments, payments and settlements across concessionaires.
- **Reporting and Analytics:** national dashboards, BI and data warehouse.
- **Central Messaging Service:** shared SMS, WhatsApp, email and USSD templates.
- **Identity and Reference Service:** national taxpayer and concession registry (parked for discussion with all stakeholders).
- **Fraud and Anomaly Monitoring, Audit and Logging, Concession Management, Security and Access Management.**
- **API Gateway and Integration:** standard APIs and event processing.

4. Integration requirements for concessionaires

- Publish transactions and assessments to CROIP as events, with idempotency keys and both event and sync timestamps (field agents may be offline).
- Conform to CROIP's canonical data model and API specification, with versioning rules.
- Authenticate with per-concessionaire credentials; each concessionaire sees only its own data.
- Pass a sandbox and conformance test before going live.

5. Trade-offs

Strengths	Weaknesses
Lower build burden on CROIP. Concessionaires keep full autonomy over operations. CROIP does not touch money, which limits regulatory exposure.	Reconciliation is harder: payments are matched after the fact across separately run systems. Ratepayers with properties in several areas need a separate login and app per concessionaire. Inconsistent payment experience and data quality across concessionaires.